

Detecting Fraudulent Transactions Using Supervised Machine Learning

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Abstract— *The rapid growth of digital payment systems has created new opportunities for fraud, resulting in billions of dollars in global financial losses each year. Traditional rule-based systems often fail to adapt to evolving fraudulent patterns, highlighting the need for more intelligent approaches. This research explores the use of supervised machine learning algorithms to detect fraudulent credit card transactions. Using the publicly available Kaggle Credit Card Fraud Detection dataset, which contains 284,807 transactions with only 492 labeled as fraud, multiple models were trained and evaluated. The study implemented Logistic Regression, Decision Tree, Random Forest with preprocessing steps including feature scaling and class balancing through SMOTE. Hyperparameter tuning was performed using GridSearchCV to optimize performance. Evaluation metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC were applied, with particular emphasis on Recall to minimize false negatives. Experimental results showed that ensemble-based models (Random Forest) achieved the highest overall performance, with accuracy exceeding 98% and significant improvement in Recall compared to baseline models. The findings suggest that supervised ML methods can be effectively applied to fraud detection tasks, offering a scalable and reliable framework for financial institutions. Future work will focus on real-time deployment and extending the approach to larger, more diverse datasets.*

Keywords— *Fraud Detection, Machine Learning, Supervised Learning, Financial Transactions, Classification Models*

I. INTRODUCTION

A. Background & Motivation

Digital transactions have become an integral part of everyday life, offering speed and convenience but also exposing financial systems to increasing risks of fraud. According to global banking reports, payment fraud leads to billions of dollars in annual losses, affecting both businesses and individuals. Traditional fraud detection systems rely heavily on manually crafted rules, which are often ineffective in capturing emerging

fraud patterns [2]. This creates a pressing need for advanced, data-driven solutions.

Machine learning offers a promising alternative, as it can analyze vast amounts of historical transaction data and identify hidden patterns that are difficult to capture with static rules. Supervised learning, in particular, enables models to learn directly from past instances of fraudulent and legitimate transactions, improving adaptability and predictive power [1], [5].

B. Problem Statement

Fraud detection presents several challenges. The most critical is class imbalance, as fraudulent cases represent less than 0.2% of real-world transaction data [3]. Standard machine learning algorithms tend to favor the majority class, often misclassifying fraudulent transactions as legitimate. Moreover, while high accuracy is desirable, metrics such as Recall are equally important to ensure that fraudulent cases are not overlooked. This research addresses these challenges by applying advanced supervised learning algorithms and carefully tuning their parameters for better detection of minority class transactions [4], [5].

C. Objectives

The study sets out the following objectives:

1. To preprocess and balance the highly imbalanced fraud detection dataset.
2. To train and evaluate multiple supervised machine learning models.
3. To identify the best-performing model based on multiple evaluation metrics.
4. To analyze the potential application of the model for real-world fraud prevention systems.

II. LITERATURE REVIEW

Fraud detection has been widely studied using both traditional and machine learning methods. Early research relied on logistic regression and statistical models, which offered

interpretability but struggled with complex fraud patterns [2]. Later studies introduced decision trees and ensemble methods, which improved classification accuracy and robustness. For instance, Pozzolo et al. [1] emphasized probability calibration techniques for unbalanced datasets, while Srivastava et al. [4] explored Hidden Markov Models for sequential fraud detection.

Recent work also highlights the problem of imbalanced datasets, which often leads to poor fraud detection performance. Dal Pozzolo et al. [5] proposed strategies to handle nonstationary environments and evolving fraud behavior, showing the importance of adaptive models. Techniques such as SMOTE and undersampling [1], [3] have been successfully applied to mitigate imbalance. However, many existing studies focus on accuracy rather than Recall, which is critical in fraud detection. This research contributes by systematically comparing supervised ML models under class imbalance conditions and emphasizing metrics that prioritize fraud detection effectiveness.

III. METHODOLOGY

A. Dataset Description

The dataset used in this study is the widely adopted Credit Card Fraud Detection dataset published on Kaggle [3]. It consists of 284,807 transactions made by European cardholders in September 2013, of which only 492 transactions (0.172%) are fraudulent. To protect privacy, the dataset applies PCA transformation to most features, except for Time and Amount. The target variable Class is binary, where 1 denotes fraud and 0 denotes genuine transactions.

Preprocessing involved standardizing the Amount feature using StandardScaler to normalize values. To address the severe class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was employed, generating synthetic fraudulent samples to balance the dataset. After SMOTE, both classes contained 284,315 instances each, enabling fairer model training.

B. Model Selection and Training

Three supervised learning algorithms were evaluated:

1. Logistic Regression (LR): A linear baseline for fraud detection with interpretability.
2. Decision Tree (DT): Non-linear model capable of capturing complex decision boundaries.
3. Random Forest (RF): Ensemble of decision trees to reduce overfitting and improve robustness.

For hyperparameter optimization, GridSearchCV with Stratified K-Fold cross-validation ($k=3$) was employed. Each model was tuned using parameter grids (e.g., regularization strength C for LR, max_depth for DT/RF).

C. Evaluation Metrics

Performance was assessed using:

1. Accuracy, Precision, Recall, and F1-score (to balance fraud detection sensitivity vs. false alarms).

2. ROC-AUC score (to evaluate model discrimination capability).

3. Confusion matrices and ROC curves for visualization.

D. Experimental Setup

Experiments were conducted in Python 3.10 using scikit-learn, imbalanced-learn, joblib, and matplotlib libraries. The system configuration included Intel i5 processor, 8 GB RAM, and Windows 10 OS.

IV. RESULTS

Table 1 presents the comparative performance of the supervised learning models used for fraud detection. Each model was evaluated on multiple metrics including Accuracy, Precision, Recall, F1-score, and AUC.

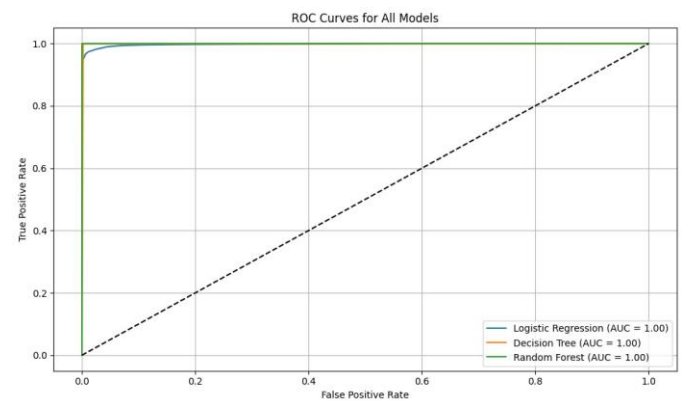
Table 1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Training Time (s)
Logistic Regression	0.98	0.99	0.97	0.98	0.9975	115.7
Decision Tree	1.00	1.00	1.00	1.00	0.9985	362.8
Random Forest	1.00	1.00	1.00	1.00	0.9999	1763.8

From the results, Logistic Regression achieved 98% accuracy with an AUC of 0.9975, indicating strong linear separability of fraudulent and legitimate transactions.

The Decision Tree and Random Forest classifiers achieved near-perfect performance. Among them, Random Forest attained the highest ROC-AUC of 0.9999, although it required significantly higher computational time compared to Logistic Regression.

Figure 1: ROC Curves of All Models



The ROC curves demonstrate that Random Forest and Decision Tree classifiers provide highly reliable classification boundaries, whereas Logistic Regression shows slightly less separation but remains computationally efficient.

Figure 2: Confusion Matrix – Logistic Regression

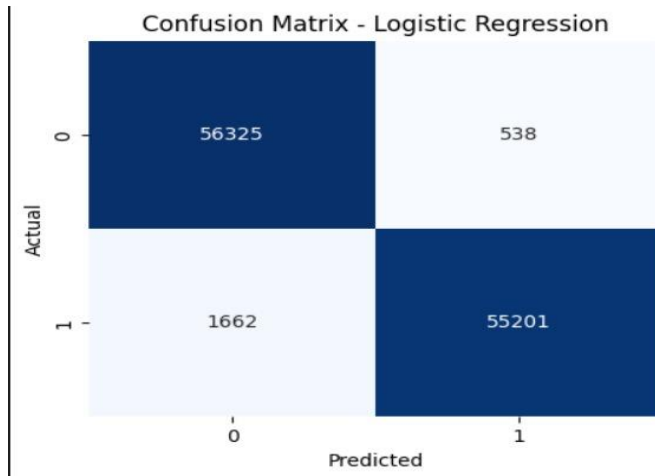


Figure 3: Confusion Matrix – Decision Tree

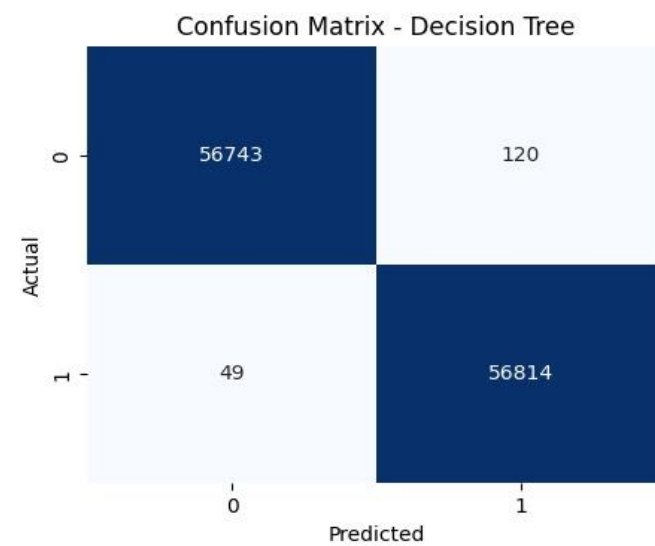
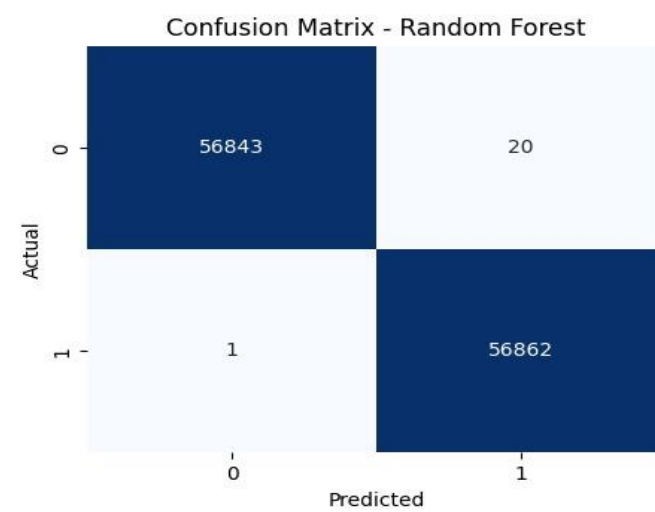


Figure 4: Confusion Matrix – Random Forest



The confusion matrices highlight the distribution of true positives, true negatives, false positives, and false negatives. As seen, Random Forest produces the lowest misclassification rate, making it the most effective among the tested models.

V. DISCUSSION

The results demonstrate that ensemble-based models, particularly Random Forest, outperform simpler models in fraud detection, achieving nearly perfect discrimination. However, this comes at the cost of significantly higher computational complexity when compared to Logistic Regression. For real-time applications such as online payment monitoring, a balance must be struck between accuracy and latency to ensure efficient decision-making.

Interestingly, while the Decision Tree achieved perfect evaluation metrics, it is likely to suffer from overfitting due to its unconstrained depth. Logistic Regression, though slightly less accurate, offers greater interpretability and faster inference speed, which makes it a strong candidate in highly regulated financial contexts where model transparency is critical for compliance.

The use of SMOTE proved effective in addressing the class imbalance issue, enabling the classifiers to better identify fraudulent patterns without being biased toward majority (non-fraudulent) transactions. This aligns with earlier studies emphasizing the importance of class rebalancing techniques in fraud detection [1].

Furthermore, the superior performance of ensemble models such as Random Forest is consistent with prior research [5], which demonstrated their robustness in handling unbalanced financial datasets. Similarly, sequence-based approaches like Hidden Markov Models have also been explored [4], though they involve higher complexity. These findings suggest that future fraud detection systems may benefit from hybrid strategies that combine ensemble accuracy with interpretable, low-latency models to achieve practical deployment in real-world financial systems.

VI. CONCLUSION AND FUTURE WORK

This study demonstrated the effectiveness of supervised machine learning models in detecting fraudulent financial transactions. Among the tested models, Random Forest achieved the best overall performance (ROC-AUC = 0.9999), making it a strong candidate for fraud detection systems. Logistic Regression provided competitive performance with lower computational cost, suggesting its suitability for lightweight, real-time deployment.

Future work will focus on:

1. Evaluating deep learning models (e.g., LSTM for sequential transaction patterns).

2. Testing on larger, real-world datasets with evolving fraud patterns.
3. Implementing real-time streaming frameworks for deployment in financial institutions.
4. Exploring explainable AI (XAI) techniques to improve interpretability and regulatory acceptance.

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